

# CNC Plasma Teaching Program

## From First Sketch to Finished Metal

School-developed NESA-aligned unit program for local review. Provisional optional Year 10, Stage 5 Industrial Technology Metal. This unit maps selected learning; it does not claim whole-course coverage or student machine competence.

10 weeks arranged as 20 proposed learning blocks, two per week. Duration, frequency, delivery year and prior learning are Teacher to confirm. Each module has three section blocks and one project or practice block; overlap and extra rehearsal are expected.

Version 1.0.0 | Prepared 7 September 2026 | Faculty review and sign-off pending | Published teaching resource. Proposed balance: 9 Site/theory, 9 Project/folio and 2 conditional Practical blocks. P means proposed block, not a confirmed timetable period.

The 2019 IND5 framework is provisional while the delivery year is unspecified. Current NSW advice starts the new syllabus with Year 9 in 2028, teaches out Year 10 on 2019 in 2028, and uses the new syllabus for all students in 2029. Confirm adoption before teaching. [Official syllabus](#) · [Implementation advice](#)

Before use: confirm timetable and syllabus, current local machine approval and configuration, authorised staff training, project/material criteria and formal assessment status. The machine stays teacher-only; students design, interpret, document and observe within the required separation.

### Selected official outcomes

| Code   | Exact official wording                                                                                                                                           |
|--------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| IND5-1 | identifies, assesses, applies and manages the risks and WHS issues associated with the use of a range of tools, equipment, materials, processes and technologies |
| IND5-2 | applies design principles in the modification, development and production of projects                                                                            |
| IND5-4 | selects, justifies and uses a range of relevant and associated materials for specific applications                                                               |
| IND5-5 | selects, interprets and applies a range of suitable communication techniques in the development, planning, production and presentation of ideas and projects     |
| IND5-6 | identifies and participates in collaborative work practices in the learning environment                                                                          |
| IND5-8 | evaluates products in terms of functional, economic, aesthetic and environmental qualities and quality of construction                                           |
| IND5-9 | describes, analyses and uses a range of current, new and emerging technologies and their various applications                                                    |

Faculty approval: \_\_\_\_\_ Date: \_\_\_\_\_ Next review: \_\_\_\_\_. Complete the delivery register after teaching; record evidence, adjustments and follow-up. Course links open the published teaching site.

# Module 1 Meet the system

Weeks 1 and 2 | Proposed blocks 1 to 4. Use flexible overlap and extra practice where required.

[Module PowerPoint](#) · [Module review and three package links](#)

| P | Type          | Focus                                     | Teaching and learning                                                                                                                                                                                      | Syllabus content                                                                                                           | Evidence / outcomes                                                                                                                                                    |
|---|---------------|-------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | Site/theory   | From idea to CNC plasma cutting           | Explain the CAD, CAM and control roles; trace one drawing through the physical system. <a href="#">Theory and illustrations</a> · <a href="#">Media or illustrated pathway</a>                             | Investigate advanced manufacturing, including CNC and plasma cutting. Compare industrial and classroom CAD/CAM production. | A reasoned system map and one corrected misconception. <a href="#">10 question knowledge check</a> · <a href="#">Scaffolded response</a><br>IND5-9                     |
| 2 | Site/theory   | Safe participation and responsibilities   | Model participation boundaries and risk controls using the illustrated source pathway. <a href="#">Theory and illustrations</a> · <a href="#">Media or illustrated pathway</a>                             | Apply risk-management principles. Demonstrate safe, collaborative workshop practices.                                      | An explanation of teacher-only operation, screening and isolation. <a href="#">10 question knowledge check</a> · <a href="#">Scaffolded response</a><br>IND5-1, IND5-6 |
| 3 | Site/theory   | Read diagrams, dimensions and coordinates | Model the square and central hole; compare dimensions, radius, diameter and two coordinate datums. <a href="#">Theory and illustrations</a> · <a href="#">Media or illustrated pathway</a>                 | Read and interpret project plans. Use specialist terminology and production records.                                       | A dimensioned explanation distinguishing size from origin. <a href="#">10 question knowledge check</a> · <a href="#">Scaffolded response</a><br>IND5-5                 |
| 4 | Project/folio | Sketch and review                         | Practise the role map and datum comparison. Sketch an intention for the offline training drawing.   <a href="#">Match the task to the system</a>   <a href="#">Which drawings describe the same shape?</a> | Produce freehand project sketches. Develop design and production folios using CAD and ICT.                                 | Annotated sketch reference and design intention; review Module 1. <a href="#">My folio</a> · <a href="#">Module review</a><br>IND5-5, IND5-6                           |

Teacher to confirm before use. Confirm cohort, prior learning, timetable and syllabus adoption before scheduling. Explain the teacher-only participation boundary before any workshop visit.

## Teaching and feedback

Every section has theory, purposeful figures, a media or illustrated pathway, 10 feedback-rich MCQs and one scaffolded response. Read feedback, return to the linked explanation and revise. Check saved reasoning; quiz completion is not operator competence.

## Module 2 Design something that can be cut

Weeks 3 and 4 | Proposed blocks 5 to 8. Use flexible overlap and extra practice where required.

[Module PowerPoint](#) · [Module review and three package links](#)

| P | Type          | Focus                                  | Teaching and learning                                                                                                                                                                                                                       | Syllabus content                                                                                                                                                                          | Evidence / outcomes                                                                                                                                             |
|---|---------------|----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5 | Project/folio | Your first dimensioned CAD sketch      | Demonstrate the offline CAD walkthrough. Students constrain the square and circle and retain the editable master. <a href="#">Theory and illustrations</a> · <a href="#">Media or illustrated pathway</a>   <a href="#">CAD walkthrough</a> | Use or modify existing designs. Develop design and production folios using CAD and ICT.                                                                                                   | Dimensions, centre relationships and named CAD revision. <b>10 question knowledge check</b> · <b>Scaffolded response</b><br>IND5-2, IND5-5, IND5-9              |
| 6 | Site/theory   | Geometry that makes a reliable outline | Trace closed contours, duplicates and retained islands; compare repair decisions using the illustrated examples. <a href="#">Theory and illustrations</a> · <a href="#">Media or illustrated pathway</a>                                    | Use or modify existing designs. Investigate advanced manufacturing, including CNC and plasma cutting.                                                                                     | A geometry diagnosis that preserves design intent. <b>10 question knowledge check</b> · <b>Scaffolded response</b><br>IND5-2, IND5-9                            |
| 7 | Project/folio | Develop a design that can be made      | Compare proposed design choices against purpose, material assumptions and checkable criteria. <a href="#">Theory and illustrations</a> · <a href="#">Media or illustrated pathway</a>                                                       | Investigate factors influencing metal design. Identify metal properties and applications; select suitable metals/profiles. Evaluate how design and work processes affect project quality. | A justified proposal with a trade-off and unresolved local decisions. <b>10 question knowledge check</b> · <b>Scaffolded response</b><br>IND5-2, IND5-4, IND5-8 |
| 8 | Project/folio | Develop and revise                     | Use contour triage and design choice activities. Revise the offline master and record what changed.   <a href="#">Find what needs checking in an outline</a>   <a href="#">Choose a revision for a clear reason</a>                         | Use or modify existing designs. Develop design and production folios using CAD and ICT. Evaluate how design and work processes affect project quality.                                    | CAD evidence, revision and improvement reason; review Module 2. <a href="#">My folio</a> · <a href="#">Module review</a><br>IND5-2, IND5-5, IND5-8              |

Teacher to confirm before use. Confirm CAD access and software versions. Keep the 100 mm square and 30 mm hole as proposed offline practice; approve any physical design, material and acceptance criteria separately.

### Access and learning support

Model one step at a time, enlarge figures and rehearse terminology. Offer paired reading, diagram annotation, spoken reasoning or teacher scribing using the same criteria. Teach dimensions, coordinates and units in context. Extend through justified revision, not machine access.

## Module 3 Prepare the production file

Weeks 5 and 6 | Proposed blocks 9 to 12. Use flexible overlap and extra practice where required.

[Module PowerPoint](#) · [Module review and three package links](#)

| P  | Type          | Focus                                | Teaching and learning                                                                                                                                                                              | Syllabus content                                                                                                                                   | Evidence / outcomes                                                                                                                                                               |
|----|---------------|--------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 9  | Project/folio | Export and check your DXF            | Model native master to DXF export and receiving-file checks; measure the known dimensions. <a href="#">Theory and illustrations</a> · <a href="#">Media or illustrated pathway</a>                 | Read and interpret project plans. Develop design and production folios using CAD and ICT.                                                          | A unit, size, contour and revision check record. <a href="#">10 question knowledge check</a> · <a href="#">Scaffolded response</a><br>IND5-5, IND5-9                              |
| 10 | Site/theory   | From drawing to toolpath in SheetCam | Read the SheetCam example: retained material, offsets, leads, sequence and machine-specific postprocessor. <a href="#">Theory and illustrations</a> · <a href="#">Media or illustrated pathway</a> | Investigate advanced manufacturing, including CNC and plasma cutting. Compare industrial and classroom CAD/CAM production.                         | An explanation of why a correct outline does not prove a safe toolpath. <a href="#">10 question knowledge check</a> · <a href="#">Scaffolded response</a><br>IND5-5, IND5-9       |
| 11 | Project/folio | Nest, name and hand over the job     | Compare layout and naming decisions, then trace one revision through the file chain. <a href="#">Theory and illustrations</a> · <a href="#">Media or illustrated pathway</a>                       | Manage a planned project sequence. Recognise material conservation and recycling. Document project management and production processes in a folio. | A clear handoff identifying quantity, revision and unresolved checks. <a href="#">10 question knowledge check</a> · <a href="#">Scaffolded response</a><br>IND5-5, IND5-6, IND5-8 |
| 12 | Project/folio | Check and hand over                  | Complete the scale check and handoff activities. Give a peer precise file-review feedback.   <a href="#">Does the drawing keep its size?</a>   <a href="#">Trace the file handoff</a>              | Manage a planned project sequence. Develop design and production folios using CAD and ICT. Demonstrate safe, collaborative workshop practices.     | Checked DXF handoff record; review Module 3. No executable TAP is required. <a href="#">My folio</a> · <a href="#">Module review</a><br>IND5-5, IND5-6                            |

Teacher to confirm before use. Confirm the receiving software and approved postprocessor before CAM output. A DXF or TAP extension alone does not establish compatibility. Students do not create executable TAP as a requirement of this sequence.

### Preparation and resources

Use each module PowerPoint and teacher notes. Prepare the offline CAD guide, training DXF, source illustrations and any approved sample. Supplier examples show their documented version and do not establish the current installation.

## Module 4 Understand teacher-controlled production

Weeks 7 and 8 | Proposed blocks 13 to 16. Use flexible overlap and extra practice where required.

[Module PowerPoint](#) · [Module review and three package links](#)

| P  | Type        | Focus                                      | Teaching and learning                                                                                                                                                                                                                                                                                          | Syllabus content                                                                                                                                                         | Evidence / outcomes                                                                                                                                                                |
|----|-------------|--------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 13 | Site/theory | Read the SureControl screen                | Read the authentic SureControl screen. Separate display, input, state and motion commands. <a href="#">Theory and illustrations</a> · <a href="#">Media or illustrated pathway</a>                                                                                                                             | Use specialist terminology and production records. Investigate advanced manufacturing, including CNC and plasma cutting.                                                 | An explanation of Set Origin versus the Origin motion command. <a href="#">10 question knowledge check</a> · <a href="#">Scaffolded response</a><br>IND5-1, IND5-5, IND5-9         |
| 14 | Site/theory | Understand the settings and cut chart      | Interpret chart labels and units. Separate software Amp from the physical power-source current check. <a href="#">Theory and illustrations</a> · <a href="#">Media or illustrated pathway</a>                                                                                                                  | Use specialist terminology and production records. Identify metal properties and applications; select suitable metals/profiles. Apply risk-management principles.        | A settings explanation that identifies missing configuration evidence. <a href="#">10 question knowledge check</a> · <a href="#">Scaffolded response</a><br>IND5-1, IND5-4, IND5-5 |
| 15 | Site/theory | Follow and observe the production sequence | Trace preflight, reference, dry run and completion using the illustrated pathway; discuss the optional clip. <a href="#">Theory and illustrations</a> · <a href="#">Media or illustrated pathway</a>                                                                                                           | Apply risk-management principles. Manage a planned project sequence. Use specialist terminology and production records.                                                  | A sequence explanation recognising that Simul commands real movement. <a href="#">10 question knowledge check</a> · <a href="#">Scaffolded response</a><br>IND5-1, IND5-5          |
| 16 | Practical   | Observe teacher production                 | If locally approved, the trained teacher demonstrates the validated sequence with required separation. Students observe from the approved location.   <a href="#">Read the label before the number</a>   <a href="#">Review the proposed working arrangement</a>   <a href="#">Teacher operating reference</a> | Apply risk-management principles. Investigate advanced manufacturing, including CNC and plasma cutting. Document project management and production processes in a folio. | Observation or illustrated-sequence evidence, clearly labelled; review Module 4. <a href="#">My folio</a> · <a href="#">Module review</a><br>IND5-1, IND5-5, IND5-6                |

Teacher to confirm before use. Before any connected-machine action, trained staff must validate current configuration, procedure, approval, extraction, access, screening and electrical isolation arrangements. Use the illustrated alternative until cleared.

### Practical decisions

Block 16 is conditional. Only a trained authorised teacher operates or commands the table, including Simul. During automatic cutting the operator is physically screened/separated and students remain outside the work area. Student access follows electrical isolation. Without approval, use the illustrated pathway and label evidence accordingly.

## Module 5 Check, improve and communicate

Weeks 9 and 10 | Proposed blocks 17 to 20. Use flexible overlap and extra practice where required.

[Module PowerPoint](#) · [Module review and three package links](#)

| P  | Type          | Focus                                        | Teaching and learning                                                                                                                                                                                                                   | Syllabus content                                                                                                                                                                    | Evidence / outcomes                                                                                                                                                                         |
|----|---------------|----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 17 | Practical     | Measure the result and recognise cut quality | Measure a teacher-released safe sample if approved. Otherwise analyse the provided dimensions and edge photographs. <a href="#">Theory and illustrations</a> · <a href="#">Media or illustrated pathway</a>                             | Measure from a project drawing accurately. Evaluate how design and work processes affect project quality.                                                                           | A measurement record separating difference, tolerance and possible cause. <a href="#">10 question knowledge check</a> · <a href="#">Scaffolded response</a><br>IND5-5, IND5-8               |
| 18 | Site/theory   | Investigate a problem using evidence         | Model an evidence-based fault report. Distinguish observations, hypotheses and the authorised teacher's next investigation. <a href="#">Theory and illustrations</a> · <a href="#">Media or illustrated pathway</a>                     | Evaluate how design and work processes affect project quality. Use specialist terminology and production records. Manage a planned project sequence.                                | A precise report without a student machine adjustment or unsupported diagnosis. <a href="#">10 question knowledge check</a> · <a href="#">Scaffolded response</a><br>IND5-1, IND5-5, IND5-8 |
| 19 | Project/folio | Evaluate and present your finished design    | Compare evidence with criteria; select an improvement and prepare captions that state what was actually completed. <a href="#">Theory and illustrations</a> · <a href="#">Media or illustrated pathway</a>                              | Evaluate how design and work processes affect project quality. Document project management and production processes in a folio. Recognise material conservation and recycling.      | A reasoned evaluation with source attribution and honest limits. <a href="#">10 question knowledge check</a> · <a href="#">Scaffolded response</a><br>IND5-5, IND5-8                        |
| 20 | Project/folio | Present and evaluate                         | Complete measurement-report and evidence-caption activities. Select and export the formative folio for teacher feedback.   <a href="#">Make the measurement report useful</a>   <a href="#">Choose evidence that supports the claim</a> | Document project management and production processes in a folio. Evaluate how design and work processes affect project quality. Demonstrate safe, collaborative workshop practices. | Readable evidence copy and one next improvement; review Module 5. <a href="#">My folio</a> · <a href="#">Module review</a><br><a href="#">Evidence guidance</a><br>IND5-5, IND5-6, IND5-8   |

Teacher to confirm before use. Release a sample only when safe to handle and identify the agreed measurement method and tolerance. Otherwise use the provided examples. Confirm the formal assessment notification and submission channel before any formal use.

### Evidence and assessment

Use the five folio cards and captions to show development. Export a readable evidence copy and keep a separate JSON backup. Formal task conditions, due dates, weightings and submission route remain Teacher to confirm.